󰏗 Phase 1: Very Detailed Theory

1. 󰏗 Installation of Linux

1.1 The Interactive Anaconda Installer (Used in RHEL/CentOS/Fedora):

- **Anaconda** is the installation program used by Red Hat-based distributions.
- It provides a **graphical** and **text-based** interface for installation.
- Supports:
  - Disk partitioning
  - Language and keyboard selection
  - Package selection

- Network configuration
- Bootloader configuration

- **Key Features**:
  - LVM and RAID setup
  - Support for kickstart scripts
  - Automated partitioning or manual mode

1.2 A Hands-Free Method of Installation (Kickstart):

- **Kickstart**: An automatic installation method used to perform unattended installations of Linux.

- Kickstart files are simple text files containing installation instructions.
- Common sections:
  - lang, keyboard, timezone
  - part, bootloader, network, user
- Can be delivered via USB, CD/DVD, or network (PXE boot)

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2. ￼ Understanding the Boot Procedure

Linux boot process has **6 major stages**:

Stage	Description
1. BIOS/UEFI	Initializes hardware (POST), loads bootloader from MBR or EFI partition
2. Boot Loader (GRUB)	Loads the kernel and initrd
3. Kernel	Initializes system hardware, mounts root filesystem, starts init
4. initrd/initramfs	Temporary root file system to load drivers/modules
5. init/systemd	First user-space process, manages services and targets
6. Runlevels/Targets	Loads default services as per target/runlevel

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3. Configuring the GRUB Boot Loader

GRUB (GRand Unified Bootloader):

- GRUB 2 is the standard bootloader used in modern Linux distros.
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Configuration File: /boot/grub2/grub.cfg (auto-generated), editable via /etc/default/grub

- Boot entries found in /etc/grub.d/
- To regenerate config:

bash

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```
grub2-mkconfig -o /boot/grub2/grub.cfg
```

Key GRUB Parameters:

- GRUB_TIMEOUT: Time to wait for selection
- GRUB_DEFAULT: Default entry

- GRUB_CMDLINE_LINUX: Kernel parameters (e.g., quiet, nomodeset)

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4. The Initial RAM Disk (initrd/initramfs)

- Temporary filesystem loaded before the actual root filesystem
- Contains essential drivers/modules (like filesystem, storage)
- Typically found as /boot/initramfs-<version>.img
- Created using:

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```
dracut -f
```

- Important during:
    - Boot time
    - Kernel upgrades

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5. Understanding Runlevels (SysVinit) and Targets (Systemd)

Traditional Runlevels:

- Defined in /etc/inittab (SysVinit systems)

Runlevel	Description
0	Halt
1	Single user
2	Multi-user (no network)
3	Multi-user with networking (no GUI)
5	Full GUI
6	Reboot

Systemd Replacements (Targets):

- /etc/systemd/system/default.target
- Common Targets:
  -

multi-user.target (Runlevel 3)

- graphical.target (Runlevel 5)

- rescue.target (Runlevel 1)

bash

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systemctl get-default systemctl set-default graphical.target

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6. 󰏗 Repository & Package Management

RedHat-based (RPM):

- **Package Format**: .rpm
- Tools: rpm, yum, dnf
- Local installation:

  bash

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  ```
  rpm -ivh package.rpm
  ```
- Repository configuration: /etc/yum.repos.d/*.repo
- Online installation:

bash

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yum install httpd dnf update

Debian-based (DEB):

- **Package Format**: .deb
- Tools: dpkg, apt
- Local install:

bash

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dpkg -i package.deb

- Online install:

  bash

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  apt update apt install apache2

Repositories:

- Store collections of packages
- Can be local or remote (official, third-party)

- Package dependencies are managed automatically by tools like yum and apt.